

Music Composition Agent | NVIDIA NIM Solution + Market Report

One-page brief covering problem fit, solution architecture, technical approach, market comparison, and execution strategy.

PROBLEM Fast editable music drafts	AI PROVIDER NVIDIA NIM only	BACKEND/API FastAPI /api/v1 on :5050	OUTPUT Chords, melody, lyrics, MIDI
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Positioning: Unlike black-box audio generators, this project creates editable musical structure: chord progressions, symbolic melody, lyrics, draft storage, and MIDI/notation export through a transparent NVIDIA NIM + FastAPI + React pipeline.

Solution Architecture & Working Flow

User Input	React UI	FastAPI	NVIDIA NIM	Validation	Storage/Export
Style, mood, theme, key, BPM, bars	Editable form and draft workspace	Routes: provider, compose, drafts, exports	Chat completion returns JSON only	Pydantic + music checks	SQLite, music21 MIDI, notation txt

Technical Approach

Generation - Prompt builder creates strict JSON schema - NIM base URL normalized to /v1 - No OpenAI/Gemini fallback	Validation + Control - Pydantic validates response shape - Music checks chord symbols and note pitches - Errors/warnings shown before export	User Execution - Edit chords, melody, lyrics in UI - Save draft through SQLite - Export MIDI via music21
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Market Landscape

Tool	How it works	Strength	Gap vs our need	Source
Suno	Prompt to full audio song	Vocals + polished demos	Less symbolic editing/control	suno.com
Udio	Prompt/lyrics to song audio	Realistic creative audio	Audio-first, less transparent	udio.com
Mubert	API generates royalty-free tracks	Good developer/API use	Background music focus	mubert.com
Boomy	Choose style, generate/publish	Beginner publishing flow	Closed platform internals	boomy.com
Our Agent	NIM -> JSON -> validate -> edit/export	Transparent, editable, testable	MVP audio quality, no vocals yet	Local project

Why Our Project Wins for Evaluation

Better for hackathon/portfolio - Shows full-stack execution, not just prompt output - Demonstrates agentic pipeline: prompt, validate, store, export - Easy to explain architecture and evaluation criteria	Better for musicians learning/composing - Outputs editable chords, melody notes, and lyrics - MIDI can be imported into DAWs for real production - Clear disclaimer and rights-review note included
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Execution depth: React + Vite frontend, FastAPI backend, NVIDIA NIM-only generation, .env security, SQLite persistence, Pydantic validation, music21 MIDI export, tests, and Docker-ready setup.
Disclaimer: generated music may resemble existing works; review before commercial use.